

TECNOLOGIES FOR INFORMATION SYSTEMS

AA 2025-26

INTRODUCTION

Prof. Letizia Tanca

<http://tanca.faculty.polimi.it/>

Dipartimento di Elettronica, Informazione e Bioingegneria
Politecnico di Milano



COURSE OVERVIEW

In this course, from the point of view of **Scientific and Methodological skills**, the students will learn how to:

- Identify the phases of Big Data Management and Analysis (including data preparation for ML), from the choice of the data sources to the production of a “data product”
- Analyze the data sources and design a Data Integration System
- Design a Data Warehouse
- Identify the Data Quality problems encountered when managing heterogeneous collections of data (with an intervention of Prof. Cappiello)

From the point of view of **Communication** (prof. Di Blas and Dr. Mataresi), the students will learn how to clearly explain a technological or methodological issue at the correct level of abstraction, **considering the common ground with the interlocutor**

Schedule at September 15th

(please always control on WeBeep whether there have been modifications)

TECHNOLOGIES FOR INFORMATION SYSTEMS a.a. 2025/26						
Prof. LETIZIA TANCA - Prospective lecture schedule						
Date	Time	Classroom	Tanca	Di Blas	Lenzi	Lecture Topic
Monday, 15 September 2025	12:15 - 14:15	25.S.1	2			Introduction, Modern Information Systems' Architectures and Big data problems
Wednesday, 17 September 2025	14:15-16.15	25.S.2	2			Introduction to Data Integration
Monday, 22 September 2025	12:15 - 14:15	25.S.1	2			Data Integration
Wednesday, 24 September 2025	14:15-16.15	25.S.2	2			Data Integration: Data Fusion problems and techniques
Monday, 29 September 2025	12:15 - 14:15	25.S.1	2			Data Integration : Semistructured data and other data formats
Wednesday, 1 October 2025	14:15-16.15	25.S.2			2	Data Integration exercises 1
Monday, 6 October 2025	12:15 - 14:15	25.S.1	2			Data Integration : Semistructured data and ontologies
Wednesday, 8 October 2025	14:15-16.15	25.S.2			2	Data Integration exercises 2
Monday, 13 October 2025	12:15 - 14:15	25.S.1			2	Data Integration exercises 3
Wednesday, 15 October 2025	14:15-16.15	25.S.2		2		Scientific Communication (exercises 1)
Monday, 20 October 2025	12:15 - 14:15	25.S.1		2		Scientific Communication (exercises 2)
Wednesday, 22 October 2025	14:15-16.15	25.S.2		2		Scientific Communication (Theory)
Monday, 27 October 2025	12:15 - 14:15	25.S.1	2			Data Quality by Prof. Cappiello
Wednesday, 29 October 2025	14:15-16.15	25.S.2			2	Data Integration exercises 4
Monday, 3 November 2025	12:15 - 14:15	25.S.1	2			Data Warehouses
Wednesday, 5 November 2025	14:15-16.15	25.S.2	2			Data Warehouses
Monday, 10 November 2025	12:15 - 14:15	25.S.1				SUSPENDED (exams)
Wednesday, 12 November 2025	14:15-16.15	25.S.2	2			Industrial Intervention (Denodo and DAMA)
Monday, 17 November 2025	12:15 - 14:15	25.S.1			2	Data integration (recap) and Data Warehouses exercises 1
Wednesday, 19 November 2025	14:15-16.15	25.S.2	2			Data Warehouses
Monday, 24 November 2025	12:15 - 14:15	25.S.1			2	Data Warehouses exercises 2
Wednesday, 26 November 2025	14:15-16.15	25.S.2			2	Data Warehouses exercises 3
Monday, 1 December 2025	12:15 - 14:15	25.S.1			2	Data Warehouses exercises 4
Wednesday, 3 December 2025	14:15-16.15	25.S.2	2			Data Warehouses, Data Lakes and Data Mesh
Monday, 8 December 2025	12:15 - 14:15	25.S.1				Immacolata Festivity
Wednesday, 10 December 2025	14:15-16.15	25.S.2				SUSPENDED (Degree day)
Monday, 15 December 2025	12:15 - 14:15	25.S.1			2	Data Warehouses exercises 5
Wednesday, 17 December 2025	14:15-16.15	25.S.2		2		Scientific Communication (Recap and exercises 3)
Monday, 22 December 2025	12:15 - 14:15	25.S.1	2			Final Lecture

IMPORTANT INFORMATION

- THE COURSE IS COMPLETELY OFFERED IN ENGLISH
- PLEASE CHECK THE SCHEDULE PUBLISHED ON THE WeBeep FOLDER BEFORE THE LECTURES, TO SEE POSSIBLE MODIFICATIONS, AND ANYWAY WATCH WeBeep FOR ALL POSSIBLE COMMUNICATIONS
- ALL THE SUPPORT MATERIALS ARE AVAILABLE ON THE WeBeep PLATFORM
- FOR THE STUDENTS THAT CANNOT ATTEND PHYSICALLY WE HAVE PROVIDED, ON WeBeep, THE RECORDINGS OF PREVIOUS YEARS' LECTURES
- THE LECTURES THAT ARE DIFFERENT FROM THE AVAILABLE RECORDINGS **SHALL BE RECORDED AND PUBLISHED**

FURTHER INFORMATION

Prerequisites:

Databases I and (possibly) Databases II (the latter can also be attended *in the same semester as TIS*).

- For the students who have not attended DB1, on WeBeep is a set of recorded lectures (**Crash Course on Databases**) that supplies the main fundamental topics.
- Also, you can resort to the book used also in the DB2 course:

P. Atzeni, S. Ceri, P. Fraternali, S. Paraboschi, R. Torlone:

***Basi di Dati (6 ed.), 6a edizione, Editore: McGraw-Hill Libri Italia,
Febbraio 2023***

or, in English:

<http://www.mcgraw-hill.co.uk/atzeni>

Exam



The whole exam IS WRITTEN and consists of:

- design exercises and questions on theoretical topics
- a **Communication Assignment** (that must be passed in order to pass the exam), in which the student has to demonstrate the acquired communication skills.

More exam details

Two main parts:

1. Theoretical Questions

Format: 3 open-ended questions.

Total Points: 15 (5 points per question).

Minimum Required Points: 7,5.

2. Exercises

Total Points: 17.

Minimum Required Points: 8,5.

The two parts can be taken **in different exam calls**, and that result will be retained (if sufficient) for the future exam sessions **of the current academic year**.

Please note: **This is a closed-book examination.**

Communication Assignment

- **Only for the Communication part** of the exam, the mark is a *pass/no pass*.
- Once the Communication part has been passed, even if the student has not passed the remaining part of the exam, this result on Communication shall be kept for the next exam calls, **and it is not necessary to do it anymore.**

Enrolment for the exam

When the students enroll for the exam, they will have the possibility to select the part or the parts that they want to take in that call

Teachers' info

- **Technical Lectures:** Prof. Letizia Tanca
 - Receiving time: Tuesdays 16:30-17:30 by previous appointment
 - Tel: 02-2399-3531
 - WeBeep of the course, containing Information for students who are interested in projects and theses (still to be updated for this academic year)
- Possibly: some **seminars** by host professors and researchers
- **Communication lectures:** Prof. Nicoletta Di Blas
- **Exercise lectures:** Dr. Emilia Lenzi

emails: name.surname@polimi.it

Proposed Lecture Time

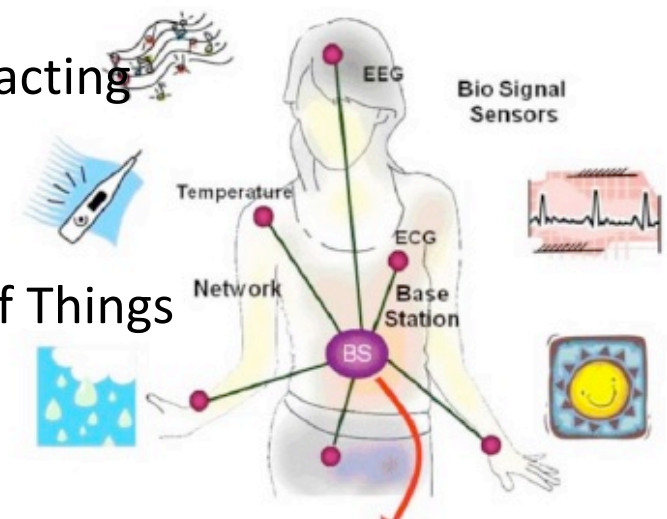
- Lectures on Mondays: 12:30 – 14.00 without interval
- Lectures on Wednesdays: 14:30 – 16.00 without interval

In case of need, sometimes we might ask you in advance to change it, e.g., anticipate it to 12:15 or 14:15

- Sometimes there might be some changes, therefore please ALWAYS CHECK THE SCHEDULE PUBLISHED ON WeBeep.

THE CHALLENGES FOR MODERN INFORMATION SYSTEMS

- ✓ Pre-history: focused on challenges that occur *within enterprises*, typically using *relational databases*
- ✓ The Web era:
 - scaling to a much larger number of data sources
 - much less structured data sources (free text, images,....).
- ✓ Nowadays:
 - large scientific experiments (e.g. in medicine) rely on data management for progress.
 - Data feed ML systems and LLMs, and train robots and autonomous agents
 - People continuously create data fragments by interacting with services on the Web.
 - Massive use of social media and smart devices
 - User-generated content merges with the Internet of Things
 - Users *as sensors and actuators*



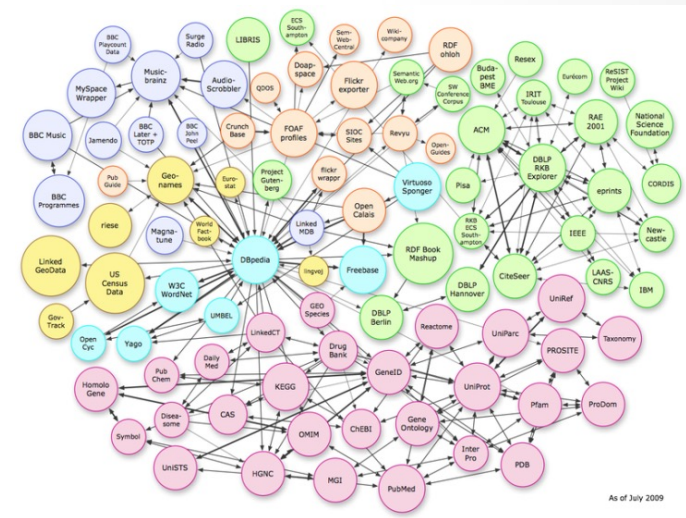
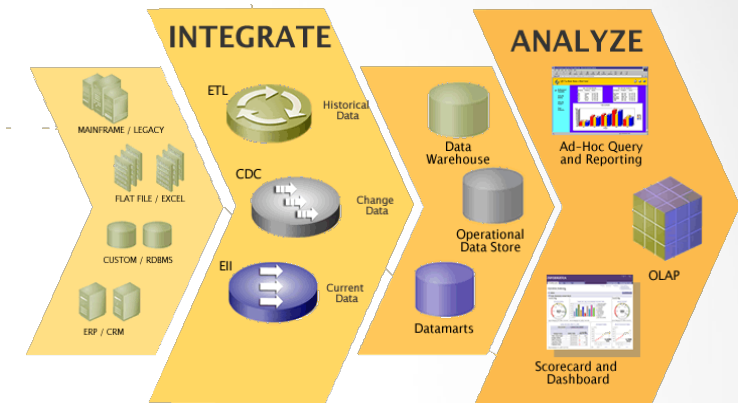
How do we control this mass of data?

The four V's of Big Data

- Volume
- Velocity
- Variety
- Veracity
- We should be able to govern:
 - data abundance
 - data and user dynamicity and mobility
 - heterogeneity, data semantics
 - incompleteness/uncertainty
 - interaction with the real-world

making sense of all this data:

→ Extract correct knowledge
and behaviours



The new challenges in the Data Management Area

What has changed in the last years:

- Technological breakthroughs in machine learning (ML) and in general in artificial intelligence (AI):
 - **Data science** as a discipline that combines elements of data cleaning and transformation, statistical analysis, data visualization, and ML techniques
 - **Automatic generation of text** (e.g., Chat GPT and other similar systems)
- Cloud computing is mainstream. The industry offers **on-demand resources** that provide on-demand, elastic storage and computation services.
- For cloud-based systems, the industry converges on **data lakes** as novel systems for **data integration** and on modern **data warehousing** query engines
- Society has become more concerned about the state of **data governance**, concentrated in data usage, e.g. **quality-aware**, privacy aware, **ethical** and **fair** use of data

The new challenges in the Data Management Area (2)

Extraction of (synthetic and useful) knowledge for decision making: build environments that mimic the progressive inspecting, observing, surveying activity with which users make decisions.

Massive data analysis and processing: A process of inspecting, cleaning, transforming, and modeling data with the goal of highlighting useful information, suggesting conclusions, and supporting decision making.

Massive use of Machine Learning, which requires clean and ethical input data to be sure that the output are reasonable and ethical

Use of reasoning services, which allow various forms of deduction and inference.

Massive data integration: People and enterprises need to integrate data and the systems that handle those data: Relational DBMSs and their extensions, legacy data and legacy DBMSs, sensors and user-generated content produce structured or unstructured data

Data warehousing: A single, complete and consistent store of data obtained from a variety of different sources made available to end users, so that they can understand and use it in a business context.[Barry Devlin]

Knowledge Distillation

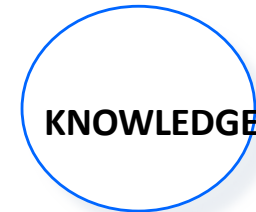
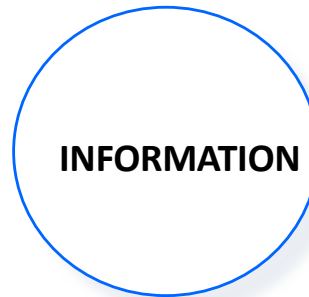
EXAMPLES:

INVOICES

SALES TREND

MARKET RULES

**STRATEGIC
DECISIONS**



VALUE ADDED

DATA ANALYSIS

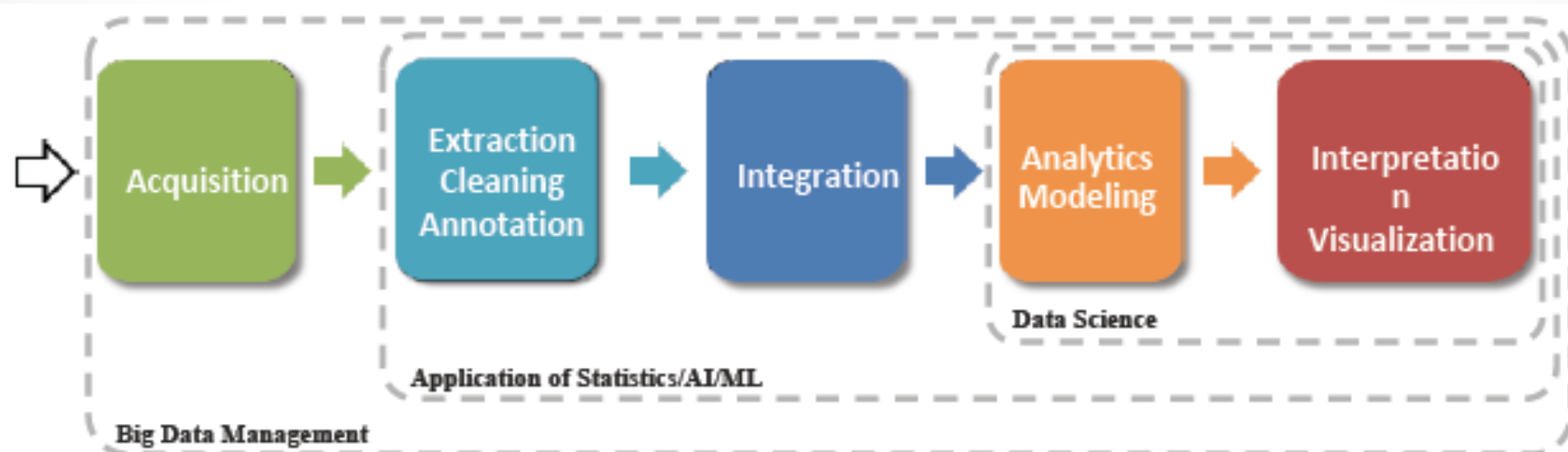
**KNOWLEDGE
DISCOVERY**

EXPERIENCE

(courtesy of prof. Fabio A. Schreiber)

The Data Pipeline: the reality behind Machine Learning and Data Analysis techniques

- The actual implementation of the Data Analysis (ML, statistics, Data Mining...) algorithm is usually less than 5% lines of code in a real, non-trivial application
- The main effort (i.e. those 95% LOC) is spent on:
 - **Data cleaning & annotation**
 - Data extraction, transformation, loading
 - **Data integration, pruning, & possibly new cleaning**
 - Parameter tuning
 - Model training & deployment
 - **THERE ARE DIFFERENT DEFINITIONS**



Credits: Beng Chin OOI – VLDB 2018 / A. Labrinidis, H. V. Jagadeesh VLDB 2012

Data Pipeline (1)

- A data pipeline is an architecture designed and built by data scientists that collects data from different sources, then stores and organizes it in a centralized data repository, such as a data warehouse. A machine learning pipeline is a workflow for designing, building and deploying a ML system.
- Both phrases use the term pipeline, but whereas a data pipeline is more of a tangible system, an ML pipeline is a theoretical series of steps. An ETL (Extract, Transform, Load) pipeline is an example of a data pipeline that extracts data from various sources, transforms it into a unified format and loads it into a destination system, usually a Data Warehouse. In Machine Learning, a data pipeline collects data and formats it into a training dataset.

Data Pipeline (2)

- A data pipeline, also known as a machine learning pipeline, is a structured workflow that automates the process of collecting, transforming, and delivering data for data analysis and machine learning tasks. It ensures data quality, consistency, and repeatability, enabling efficient and reliable analysis and model building.

Key Components of a Data Pipeline:

1. Data Ingestion:

This involves collecting data from various sources, which could be databases, APIs, files, or other systems.

2. Data Transformation:

Data is cleaned, processed, and transformed to a suitable format for analysis or model training. This may include tasks like data cleaning, feature engineering, and data standardization.

3. Data Storage:

The transformed data is stored in a central repository, such as a data warehouse or data lake, for analysis and model building.

4. Model Training and Evaluation:

The data pipeline is used to train machine learning models, evaluate their performance, and deploy them for prediction.

5. Model Deployment:

The trained models are deployed into a production environment where they can make predictions on new data.

Benefits of Using Data Pipelines

- **Improved Data Quality:**
 - Data pipelines ensure that data is cleaned, transformed, and validated, leading to more reliable and accurate insights and model predictions.
- **Automation and Efficiency:**
 - By automating the data processing workflow, data pipelines save time and effort, allowing data scientists and engineers to focus on more strategic tasks.
- **Consistency and Repeatability:**
 - Data pipelines provide a standardized approach to data processing, ensuring consistent results and facilitating easier maintenance and upgrades.
- **Scalability:**
 - Data pipelines can be scaled to handle large volumes of data, allowing for the efficient processing of big data sets.
- **Enhanced Collaboration:**
 - Data pipelines enable better communication and collaboration among data scientists, engineers, and other stakeholders.

Example

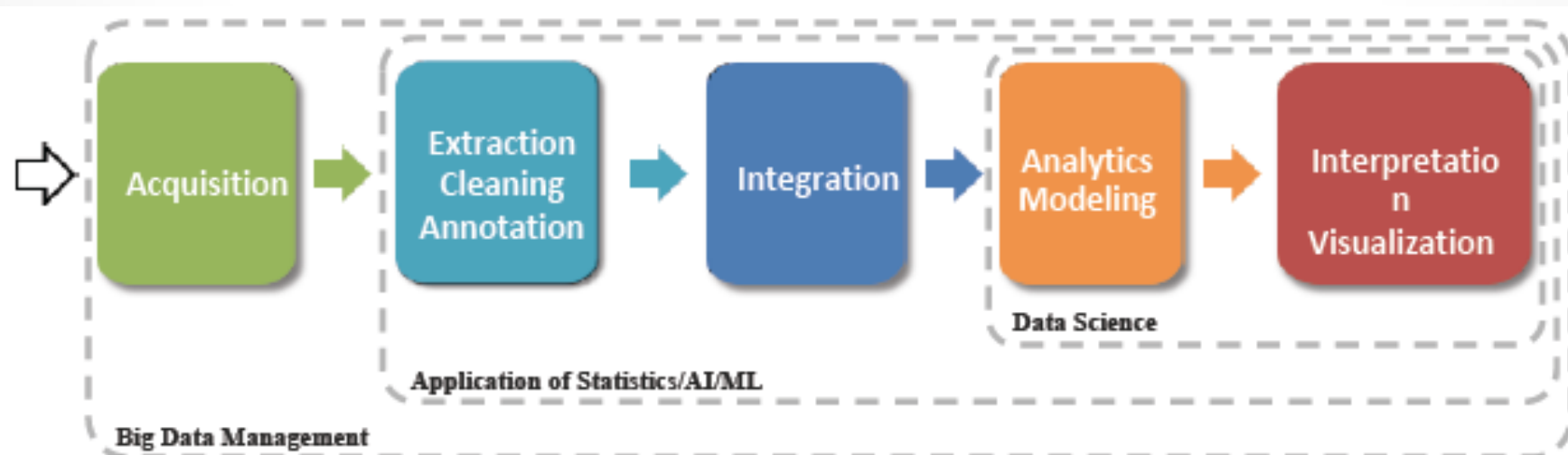
Consider a machine learning pipeline for **predicting customer churn**. The pipeline might start with data collection from a customer relationship management (CRM) system and a mobile app. Data transformation steps might involve cleaning, feature engineering (e.g., creating a "days since last purchase" feature), and data standardization. The transformed data would then be stored in a data warehouse. A model would be trained on the data, and its performance evaluated. Finally, the trained model could be deployed to make predictions on new customer data, identifying those at risk of churning.

Knowledge-extraction Lifecycle

- Source selection
- Data integration
- Various methods for knowledge extraction:
 - Querying
 - Data analysis:
 - Statistics
 - Data warehousing
 - Data Mining
 - Machine Learning
 - Visualization

The Data Pipeline: Recall

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 - Model training & deployment
 - ...



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Conclusion

In this course, we concentrate on two main aspects of data manipulation and usage:

- **Data Integration**, a delicate operation where we try to extract and **reconcile** data from different data sources
- **Data Warehousing**, where we learn how we can design and use data repositories that **collect historical data for future strategical decisions**



- The last lectures will be devoted to the **advances on these topics**